

PathFinder — AI Documentation

1. AI Overview

PathFinder integrates artificial intelligence across multiple features to enhance the user experience for both students and organizations. The AI system is designed with a **dual-mode architecture**: it can operate with a live external LLM API for production-quality responses, or fall back to intelligent mock responses when no API key is configured.

This ensures the platform remains fully functional in development environments, demo scenarios, and situations where API quotas are exhausted.

2. AI Architecture

The AI layer is implemented entirely in the **backend** (`backend/routes/ai.ts`) to ensure:

- **Security:** API keys are never exposed to the client.
- **Consistency:** All AI processing uses the same pipeline regardless of the frontend client.
- **Auditability:** Every AI generation is logged in the `AIGeneration` database table.

Technology Stack

- **Primary AI Provider:** Google Generative AI (Gemini 1.5 Flash model)
- **PDF Processing:** `pdf-parse` library for extracting text from uploaded resume PDFs
- **Fallback System:** Deterministic mock responses using student profile data

Dual-Mode Operation

```
if (process.env.GEMINI_API_KEY) {  
  // Live mode: Send structured prompts to Gemini API  
  // Parse JSON responses, extract scores and text  
} else {  
  // Mock mode: Generate contextual responses using profile data  
  // Simulate processing delay for realistic UX  
}
```

3. Resume Analysis (PDF Parsing)

Endpoint: `POST /api/ai/parse-resume`

This feature allows students to upload a PDF resume and have the system automatically extract structured profile data.

Workflow

1. Student uploads a PDF file via multipart form data.
2. Backend receives the file buffer (stored in memory, not on disk).
3. `pdf-parse` library extracts raw text from the PDF.
4. Text is sent to the AI model with a structured prompt requesting JSON output.
5. AI returns a JSON object with: `firstName`, `lastName`, `bio`, `major`, `university`, `graduationYear`, and `skills` array.
6. Frontend pre-fills the student profile form with extracted data.

Mock Fallback

When no API key is available, the system returns a predefined mock profile with simulated 1.5-second delay to mimic API processing time.

4. Smart Match Scoring Engine

Endpoint: `POST /api/applications` (triggered automatically on application submission)

The match scoring engine evaluates how well a student's profile aligns with an opportunity's requirements.

Live Mode (with API key)

1. A structured prompt is sent to the AI containing:
 - Opportunity title and full description
 - Student bio, skills list, and cover letter
2. AI returns a JSON object: { matchScore: number, matchReason: string }
3. Score ranges from 0-100 representing percentage compatibility.
4. Reason provides a one-sentence human-readable explanation.

Mock Mode (without API key)

1. System generates a random score between 60-95.
2. Reason is contextually generated based on the student's skills:
 - Score >= 85: "Excellent match" messaging with skill names
 - Score 70-84: "Solid candidate" messaging
 - Score < 70: "Shows potential" messaging

Asynchronous Processing

Match scoring runs as a background async operation. The application submission returns immediately (201 status) while the score is calculated and updated in the database afterward. This ensures the student is not blocked by AI processing time.

5. Document Generation Suite

PathFinder includes four AI-powered document generators, each personalized to the student's profile and (optionally) a specific opportunity.

5.1 Statement of Purpose (SOP) Generator

Endpoint: POST /api/ai/generate-sop

Generates a formal SOP incorporating:

- Student's name, university, major, and graduation year
- Skills and certifications
- Bio and additional context provided by the student
- Opportunity title and organization name

5.2 Cover Letter Generator

Endpoint: POST /api/ai/generate-cover-letter

Generates a professional cover letter with:

- Proper business letter formatting
- Skills alignment with opportunity requirements
- Eligibility acknowledgment (if specified)

5.3 Essay Generator

Endpoint: POST /api/ai/generate-essay

Generates an academic essay with:

- Custom topic prompt or auto-generated topic from opportunity
- Integration of academic background and certifications
- Forward-looking conclusion

5.4 Personal Statement Generator

Endpoint: POST /api/ai/generate-personal-statement

Generates a comprehensive personal statement with:

- Narrative arc from early curiosity to current expertise
- Skills and certifications woven into the story
- Vision for future impact

Generation Storage

All generated documents are saved to the AIGeneration table with:

- User ID (who generated it)
 - Type (SOP, COVER_LETTER, ESSAY, PERSONAL_STATEMENT)
 - Prompt (input context)
 - Output (full generated text)
 - Opportunity ID (if linked to a specific opportunity)
-

6. Profile Strength Analysis

Endpoint: GET /api/ai/profile-strength

Evaluates the completeness of a student's profile across 11 checkpoints:

#	Checkpoint	Category
1	First Name filled	Basic
2	Last Name filled	Basic
3	University filled	Basic
4	Major filled	Basic
5	Bio written	Basic
6	Graduation Year set	Basic
7	At least 3 skills added	Skills
8	At least 5 skills added	Skills
9	Resume uploaded	Documents
10	At least 1 certification	Certifications
11	At least 1 portfolio link	Links

Returns a percentage score and the detailed checklist for the frontend to display.

7. Opportunity Recommendations

Endpoint: GET /api/ai/recommendations

Provides personalized opportunity suggestions using a keyword-matching algorithm:

Scoring Algorithm

1. Fetch the student's skills as lowercase strings.
 2. Fetch the 50 most recent published opportunities.
 3. For each opportunity, concatenate title + description + eligibility + category into a searchable text block.
 4. Award points:
 - +20 points per skill match found in the text
 - +15 points if student's major appears in the text
 - +10 points if the opportunity is featured
 - +5 points if the deadline is still active
 5. Cap score at 98.
 6. Assign minimum random score (15-40) if no matches found.
 7. Sort by score descending and return top 10.
-

8. Candidate Summary (Organization Feature)

Endpoint: GET /api/ai/candidate-summary/:applicationId

Generates a structured profile summary for organizations reviewing applicants:

- Student name, university, major, graduation year
 - Skills list and certifications
 - Document inventory and portfolio links
 - Quality assessment checkmarks (resume uploaded, bio provided, certifications present, cover letter submitted)
-

9. Limitations

1. **Context Window:** The AI model has token limits. Very long resumes or descriptions may be truncated.
2. **Accuracy:** AI-generated match scores are heuristic-based and should be used as guidance, not definitive assessments.
3. **Mock Mode:** Without an API key, document quality is template-based rather than dynamically generated.

4. **Language:** Currently supports English-language content only.
 5. **Rate Limits:** External AI APIs may impose rate limits during high-usage periods.
-

10. Future Improvements

- Integration with multiple AI providers (OpenAI, Anthropic) for redundancy
 - Fine-tuned models specifically trained on academic application documents
 - AI interview preparation module with practice questions
 - Plagiarism detection for submitted application materials
 - Multi-language document generation
-

End of AI Documentation